More On Linear Regression Models

— Geometry and Orthogonality

STAT 24510-30040

(Winter 2025)

Review vector-matrix notation for linear model

Recall that the linear regression model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_r x_{ir} + \varepsilon_i, \qquad \varepsilon_i \;\; i.i.d. \sim N(0, \sigma^2), \quad i = 1, \cdots, n$$

represents a set of n model equations, each for one observation. By conversion, p denotes the number of parameters β_i in the model. Here $p = r + 1$. We consider the case $p < n$. Writing the equations in vector-matrix form,

$$\underset{[n\times 1]}{\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ \vdots \\ Y_n \end{bmatrix}} = \underset{[n\times p]}{\begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1r} \\ 1 & x_{21} & x_{22} & \cdots & x_{2r} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nr} \end{bmatrix}} \underset{[p\times 1]}{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_r \end{bmatrix}} + \underset{[n\times 1]}{\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \vdots \\ \varepsilon_n \end{bmatrix}}$$

The compact vector-matrix notation becomes

$$Y = X\beta + \varepsilon, \qquad \varepsilon \sim N_n\left(0_n, \sigma^2 I_n\right)$$

We have shown that, the MLE of the p dimensional parameter vector β is the same as the LSE obtained by minimizing the residual sum of squares

$$\sum_{i=1}^{n} [Y_i - (\beta_0 + \beta_1 x_{i1} + \cdots + \beta_r x_{ir})]^2 = \|Y - X\beta\|^2$$

The minimum can be derived by setting

$$\frac{d}{d\beta}\|Y - X\beta\|^2 = 0_p$$

Vector-matrix differentiation leads to the **normal equation**

$$(X^T X)\hat{\beta} = X^T Y$$

When $X^T X$ is invertible, that is, when

$$rank(X) = p, \qquad (p < n)$$

the estimator of β can be expressed as

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

We have shown that the estimator is unbiased.

$$\mathbb{E}(\hat{\beta}) = (X^T X)^{-1} X^T \mathbb{E}(Y) = (X^T X)^{-1} X^T \mathbb{E}(X\beta + \varepsilon) = \beta$$

We have also derived

$$Cov(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$$

where $Cov(\hat{\beta})$ is the variance-covariance matrix of $\hat{\beta}$,

$$Cov(\hat{\beta}) = \begin{bmatrix} var(\hat{\beta}_0) & cov(\hat{\beta}_0, \hat{\beta}_1) & \cdots & cov(\hat{\beta}_0, \hat{\beta}_r) \\ cov(\hat{\beta}_1, \hat{\beta}_0) & var(\hat{\beta}_1) & \cdots & cov(\hat{\beta}_1, \hat{\beta}_r) \\ \vdots & \vdots & \ddots & \vdots \\ cov(\hat{\beta}_r, \hat{\beta}_0) & \cdots & \cdots & var(\hat{\beta}_r) \end{bmatrix}$$

Consequently, the $i.i.d.$ normality assumption leads to the normality of the estimator.

$$\hat{\beta} \;\sim\; N_p\Big(\;\beta,\; \sigma^2 (X^T X)^{-1}\;\Big)$$

The MLE for the variance parameter σ^2 is

$$\hat{\sigma}^2_{ML} = \frac{RSS}{n}$$

where we used the conventional notation

$$RSS = \sum_{i=1}^{n} \Big(Y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots \hat{\beta}_2 x_{i,r})\Big)^2 = \|Y - X\hat{\beta}\|^2$$

We have proved that

$$\frac{RSS}{\sigma^2} \sim \chi^2_{n-p}$$

Therefore, an unbiased, commonly used estimator for the variance parameter σ^2 is

$$\hat{\sigma}^2 = \frac{RSS}{n-p}$$

Special case of $p = 2$

In the case of simple linear regression model corresponding to $p = 2$, the design matrix has the simple form

$$X = \begin{bmatrix} 1 & x_{11} \\ 1 & x_{21} \\ \vdots & \vdots \\ 1 & x_{n1} \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}$$

We can write out the entries in the matrix explicitly.

$$X^T X = \begin{bmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{bmatrix} = n \begin{bmatrix} 1 & \bar{x} \\ \bar{x} & \frac{1}{n}\sum_{i=1}^n x_i^2 \end{bmatrix}$$

To obtain the inverse matrix, first find the determinant, which has several useful expressions,

$$det(X^TX) = n\sum_{i=1}^n x_i^2 - \Big(\sum_{i=1}^n x_i\Big)^2 = n\left(\sum_{i=1}^n x_i^2 - n\bar{x}^2\right) = n\sum_{i=1}^n (x_i - \bar{x})^2$$

Using the inverse matrix relation

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

we obtain an expression of $(X^TX)^{-1}$,

$$(X^TX)^{-1} = \frac{1}{\sum_{i=1}^n (x_i-\bar{x})^2}\begin{bmatrix} \frac{1}{n}\sum_{i=1}^n x_i^2 & -\bar{x} \\ & \\ -\bar{x} & 1 \end{bmatrix}$$

For the sake of comparison and confirmation, we can check $\hat{\beta}$ in the matrix form. First, we note that

$$X^TY = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n y_i \\ \\ \sum_{i=1}^n x_i y_i \end{bmatrix}$$

Now we check the matrix form solution of $\hat{\beta}$,

$$\begin{aligned}
\hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} &= (X^TX)^{-1}X^TY \\
&= \frac{1}{\sum_{i=1}^n (x_i-\bar{x})^2}\begin{bmatrix} \frac{1}{n}\sum_{i=1}^n x_i^2 & -\bar{x} \\ & \\ -\bar{x} & 1 \end{bmatrix}\begin{bmatrix} \sum_{i=1}^n y_i \\ \\ \sum_{i=1}^n x_i y_i \end{bmatrix} \\
&= \frac{1}{\sum_{i=1}^n (x_i-\bar{x})^2}\begin{bmatrix} \sum_{i=1}^n x_i^2\bar{y} - \bar{x}\sum_{i=1}^n x_i y_i \\ \\ -n\bar{x}\bar{y} + \sum_{i=1}^n x_i y_i \end{bmatrix} \\
&= \frac{1}{\sum_{i=1}^n (x_i-\bar{x})^2}\begin{bmatrix} \sum_{i=1}^n (x_i-\bar{x})^2\bar{y} + n\bar{x}^2\bar{y} - \bar{x}\sum_{i=1}^n x_i y_i \\ \\ -n\bar{x}\bar{y} + \sum_{i=1}^n x_i y_i \end{bmatrix} \\
&= \begin{bmatrix} \bar{y} - \bar{x}\,\dfrac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n (x_i-\bar{x})^2} \\ \\ \dfrac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n (x_i-\bar{x})^2} \end{bmatrix}
\end{aligned}$$

The result agrees with our original derivation before using matrix notation,

$$\begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} = \begin{bmatrix} \bar{y} - \bar{x}\,\hat{\beta}_1 \\ \\ \dfrac{\sum_{i=1}^n (x_i-\bar{x})(y_i-\bar{y})}{\sum_{i=1}^n (x_i-\bar{x})^2} \end{bmatrix}$$

The covariance matrix has simple entries,

$$Cov(\hat{\beta}) = Cov\begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} = \sigma^2(X^TX)^{-1} = \frac{\sigma^2}{\sum_{i=1}^n (x_i-\bar{x})^2}\begin{bmatrix} \frac{1}{n}\sum_{i=1}^n x_i^2 & -\bar{x} \\ & \\ -\bar{x} & 1 \end{bmatrix}$$

In this simple care, we have explicit and intuitive expressions for the variance and covariance, as we derived before.

$$\begin{aligned}
var(\hat{\beta}_1) &= \frac{\sigma^2}{\sum_{i=1}^n (x_i-\bar{x})^2} \\
var(\hat{\beta}_0) &= \frac{\sigma^2\sum_{i=1}^n x_i^2}{n\sum_{i=1}^n (x_i-\bar{x})^2} = \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i-\bar{x})^2}\right) \\
cov(\hat{\beta}_0, \hat{\beta}_1) &= \frac{-\sigma^2\bar{x}}{\sum_{i=1}^n (x_i-\bar{x})^2}
\end{aligned}$$

Inference on model coefficients

The formula of variance-covariance matrix for the coefficient vector estimate $Cov(\hat{\beta}) = \sigma^2(X^TX)^{-1}$ allows us to conduct statistical inference on the parameters, with an estimator for σ^2.

For example, for each parameter β_i, $i = 0, \cdots, p-1$, a t-test is commonly conducted to compare the current model and the model removing predictor x_i, keeping all other predictors.

$$\begin{cases} H_o: & \beta_i = 0 \text{ (in the current model)} \\ H_a: & \beta_i \neq 0 \text{ (in the current model)} \end{cases}$$

The test statistic is

$$\frac{\hat{\beta}_i - 0}{s.e.(\hat{\beta}_i)}$$

In the denominator, the standard error is

$$s.e.(\hat{\beta}_i) = \sqrt{\widehat{var}(\hat{\beta}_i)}$$

where

$$var(\hat{\beta}_i) = \text{ the } i\text{th diagonal element of } Cov(\hat{\beta}) = \sigma^2(X^TX)^{-1}$$

and $\widehat{var}(\hat{\beta}_i)$ is the estimated $var(\hat{\beta}_i)$ with σ^2 replaced by an estimator $\hat{\sigma}^2$.

Commonly, the d.f.-normalized RSS of the model

$$\hat{\sigma}^2 = \frac{\|Y - X\hat{\beta}\|^2}{n-p} = \frac{RSS}{n-p}$$

is used as the (unbiased) estimator of σ^2.

We derived the result

$$\frac{RSS}{\sigma^2} \sim \chi^2_{n-p}$$

and the independence relation

$$RSS \perp\!\!\!\perp \hat{\beta}$$

In addition, under the null of $H_o: \beta_1 = \cdots = \beta_{p-1} = 0$, we have

$$\frac{MSS}{\sigma^2} \sim \chi^2_{p-1}, \qquad MSS = SS_{model} = \|\hat{Y} - \bar{Y}\|$$

Consequently we can conduct F test for the null, since

$$F = \frac{SS_{model}/(p-1)}{SS_e/(n-p)} = \frac{(RSS_{null} - RSS_{full})/((n-1)-(n-p))}{RSS_{full}/(n-p)} \sim F_{p-1,n-p} \qquad under\ H_o$$

We reject the null when the ratio is too large, when $F > F_{p-1,n-p,1-\alpha}$ for a given α.

We may conduct t tests for a variety of hypotheses on the β_i's.

For example, we ca obtain a null distribution for the hypothesis test,

$$\frac{\hat{\beta}_i - 0}{s.e.(\hat{\beta}_i)} \sim t_{n-p} \qquad under \quad H_o: \beta_i = 0 \text{ (in the current model)}$$

Geometry of least squares and the projection matrix

Recall that, the LS model estimated response, a.k.a. the fitted values, are denoted as $\hat{Y}$,

$$\hat{Y} = X\hat{\beta} = X(X^TX)^{-1}X^TY = P_XY = HY$$

where we used the notations

$$H = X(X^TX)^{-1}X^T = P_X$$

The conventional notation "hat matrix" H is from the relation that HY is the model fitted Y,

$$HY = \hat{Y}$$

and the notation

$$P_X = X(X^TX)^{-1}X^T$$

is to emphasize that P_X is a projection matrix with respect to the column space of X.
(In this course, projection means orthogonal projection, unless stated otherwise.)

$$Span\{X\} = \text{Column space of } X = \{\text{All linear combinations of columns of } X\} = \{X\beta, \beta \in \mathbb{R}^p\}$$

In the following, we summarize properties of the projection matrix P_X and the implications in linear regression models.

<u>Projection matrix P_X</u> $(rank(X) = p < n)$

- P_X is symmetric: $P_X^T = P_X$.
- P_X is idempotent: $P_X^2 = P_X$.
- P_XY is in the column space of X.
- Eigenvalues of P_X are 1's and 0's, based on the idempotent property.

- $Rank(P_X) = rank(X^TX) = p$.
- The above properties imply that, of the n eigenvalues $\lambda_1, \cdots, \lambda_n$ of P_X, we may order them as

$$\lambda_1 = \cdots = \lambda_p = 1, \qquad \lambda_{p+1}, \cdots, \lambda_n = 0$$

<u>The complement projection matrix $I_n - P_X$</u>

$I_n - P_X$ is also a projection matrix, where I_n is the $n \times n$ identity matrix.

- $I_n - P_X$ is symmetric. $(I_n - P_X)^T = I_n - P_X$.
- $I_n - P_X$ is idempotent. $(I_n - P_X)^2 = I_n - P_X$
- The residual vector $\varepsilon = (I_n - P_X)Y$ is in the orthogonal complement of the column space of X.
- Eigenvalues of $I_n - P_X$ are 1's and 0's.
- $Rank(I_n - P_X) = n - p$

<u>Projection matrix relations</u>

The projection matrices have the relation

$$P_X(I_n - P_X) = (I_n - P_X)P_X = 0_{n\times n}$$

We may write this orthogonality as

$$P_X \perp I_n - P_X$$

When applied to Y,

$$Y = P_XY + (I_n - P_X)Y$$

and

$$\|Y\|^2 = \|P_XY\|^2 + \|(I - P_X)Y\|^2$$

The pythagorean rule corresponds to the partition of sums of squares. The last term is the model RSS since the residual vector

$$\hat{\varepsilon} = Y - \hat{Y} = (I_n - P_X)Y = (I_n - P_X)\varepsilon$$

Correlation of explanatory variables and effects on parameter estimation

We investigate lower dimensional cases to illustrate the projection properties of least square methods in linear regression model.

<u>Example 1</u>

Parameter dimension $p = 1$, no intercept.

$$Y_i = \beta_1 x_i + \varepsilon_i, \qquad \varepsilon_i \ \ i.i.d. \sim N(0, \sigma^2), \quad i = 1, \cdots, n$$

The design matrix of the linear model is

$$X = [\boldsymbol{x}] = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Then the hat matrix $H = P_X$ has the form

$$P_X = P_{\boldsymbol{x}} = \boldsymbol{x}\left(\boldsymbol{x}^T\boldsymbol{x}\right)^{-1}\boldsymbol{x}^T = \frac{\boldsymbol{x}\boldsymbol{x}^T}{\boldsymbol{x}^T\boldsymbol{x}} = \frac{\boldsymbol{x}\boldsymbol{x}^T}{\|\boldsymbol{x}\|^2}$$

Example 2a

Parameter dimension $p = 2$, no intercept.

$$Y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i, \qquad \varepsilon_i \;\; i.i.d. \sim N(0, \sigma^2), \quad i = 1, \cdots, n$$

The design matrix of the linear model is

$$X = [\boldsymbol{x}_1 \; \boldsymbol{x}_2] = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \\ \vdots & \vdots \\ x_{n1} & x_{n2} \end{bmatrix}$$

and

$$X^T X = \begin{bmatrix} \boldsymbol{x}_1^T \\ \boldsymbol{x}_2^T \end{bmatrix} [\boldsymbol{x}_1 \; \boldsymbol{x}_2] = \begin{bmatrix} \boldsymbol{x}_1^T\boldsymbol{x}_1 & \boldsymbol{x}_1^T\boldsymbol{x}_2 \\ \boldsymbol{x}_2^T\boldsymbol{x}_1 & \boldsymbol{x}_2^T\boldsymbol{x}_2 \end{bmatrix}$$

Example 2b

Parameter dimension $p = 2$, no intercept, orthogonal subspaces.

$$Y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i, \qquad \varepsilon_i \;\; i.i.d. \sim N(0, \sigma^2), \quad i = 1, \cdots, n$$

In addition, the columns of $X = [\boldsymbol{x}_1 \; \boldsymbol{x}_2] = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \\ \vdots & \vdots \\ x_{n1} & x_{n2} \end{bmatrix}$ are orthogonal,

$$\boldsymbol{x}_1 \perp \boldsymbol{x}_2 \qquad \Leftrightarrow \qquad \boldsymbol{x}_1^T\boldsymbol{x}_2 = 0$$

Then

$$X^T X = \begin{bmatrix} \boldsymbol{x}_1^T\boldsymbol{x}_1 & 0 \\ 0 & \boldsymbol{x}_2^T\boldsymbol{x}_2 \end{bmatrix} = \begin{bmatrix} \|\boldsymbol{x}_1\|^2 & 0 \\ 0 & \|\boldsymbol{x}_2\|^2 \end{bmatrix}$$

Its inverse matrix is

$$(X^T X)^{-1} = \begin{bmatrix} \frac{1}{\|\boldsymbol{x}_1\|^2} & 0 \\ 0 & \frac{1}{\|\boldsymbol{x}_2\|^2} \end{bmatrix}$$

Then the hat matrix of the linear model $H = P_X$ can be written as

$$P_X = X(X^TX)^{-1}X = [\boldsymbol{x}_1 \; \boldsymbol{x}_2] \begin{bmatrix} \frac{1}{\|\boldsymbol{x}_1\|^2} & 0 \\ 0 & \frac{1}{\|\boldsymbol{x}_2\|^2} \end{bmatrix} \begin{bmatrix} \boldsymbol{x}_1^T \\ \boldsymbol{x}_2^T \end{bmatrix} = \frac{\boldsymbol{x}_1{\boldsymbol{x}_1}^T}{\|\boldsymbol{x}_1\|^2} + \frac{\boldsymbol{x}_2{\boldsymbol{x}_2}^T}{\|\boldsymbol{x}_2\|^2}$$

We have derived the decomposition

$$P_X = P_{\boldsymbol{x}_1} + P_{\boldsymbol{x}_2}$$

Therefore

$$P_X Y = P_{\boldsymbol{x}_1} Y + P_{\boldsymbol{x}_2} Y$$

By the orthogonal property,

$$P_{\boldsymbol{x}_1}^T P_{\boldsymbol{x}_2} = \frac{(\boldsymbol{x}_1{\boldsymbol{x}_1}^T)^T(\boldsymbol{x}_2{\boldsymbol{x}_2}^T)}{\|\boldsymbol{x}_1\|^2\|\boldsymbol{x}_2\|^2} = \frac{\boldsymbol{x}_1({\boldsymbol{x}_1}^T\boldsymbol{x}_2){\boldsymbol{x}_2}^T}{\|\boldsymbol{x}_1\|^2\|\boldsymbol{x}_2\|^2} = 0_{n\times n}, \qquad \text{analogously}, \qquad P_{\boldsymbol{x}_2}^T P_{\boldsymbol{x}_1} = 0_{n\times n}.$$

We can write the orthogonality as

$$P_{\boldsymbol{x}_1} \perp P_{\boldsymbol{x}_2}$$

Then

$$\begin{aligned} \|P_X Y\|^2 &= (P_{\boldsymbol{x}_1} Y + P_{\boldsymbol{x}_2} Y)^T (P_{\boldsymbol{x}_1} Y + P_{\boldsymbol{x}_2} Y) \\ &= Y^T P_{\boldsymbol{x}_1}^T P_{\boldsymbol{x}_1} Y + Y^T P_{\boldsymbol{x}_2}^T P_{\boldsymbol{x}_2} Y + 0 \\ &= (P_{\boldsymbol{x}_1} Y)^T P_{\boldsymbol{x}_1} Y + (P_{\boldsymbol{x}_2} Y)^T P_{\boldsymbol{x}_2} Y = \|P_{\boldsymbol{x}_1} Y\|^2 + \|P_{\boldsymbol{x}_2} Y\|^2 \end{aligned}$$

The sc anova SS equation

$$\|Y\|^2 = \|P_X Y\|^2 + \|(I - P_X)Y\|^2$$

has the decomposition

$$\|Y\|^2 = \|P_{\boldsymbol{x}_1} Y\|^2 + \|P_{\boldsymbol{x}_2} Y\|^2 + \|(I - P_X)Y\|^2$$

Remarks

- The decomposition corresponds to the partition of sums of squares. The equation shows that the projection order of $\boldsymbol{x}_1, \boldsymbol{x}_2$ does not affect the pythagorean relationship in the SS decomposition.

- Note that in Example 2a, such pythagorean relationship in the SS decomposition does not hold without the $\boldsymbol{x}_1^T\boldsymbol{x}_2 = 0$ assumption.

- To recap, if

  $$\boldsymbol{x}_1 \perp \boldsymbol{x}_2$$

  then

  - ANOVA is unaffected by order.
  - Estimated regression coefficients are the same as when Y is regressed on each $\boldsymbol{x}_i$ individually and separately (in the no-intercept case).
  - The variance-covariance matrix of the estimated coefficients

    $$Cov(\hat{\boldsymbol{\beta}}) = Cov \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} = \sigma^2 (X^T X)^{-1} = \sigma^2 \begin{bmatrix} \frac{1}{\|\boldsymbol{x}_1\|^2} & 0 \\ 0 & \frac{1}{\|\boldsymbol{x}_2\|^2} \end{bmatrix}$$

  - Consequently,

    $$corr\left(\hat{\beta}_0, \hat{\beta}_1\right) = 0$$

Example 2c

A concrete example of parameter dimension $p = 2$, no intercept, orthogonal subspaces.

$$\begin{bmatrix} Y_{11} \\ Y_{12} \\ Y_{13} \\ Y_{14} \\ Y_{21} \\ Y_{22} \\ Y_{23} \\ Y_{24} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} + \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{12} \\ \varepsilon_{13} \\ \varepsilon_{14} \\ \varepsilon_{21} \\ \varepsilon_{22} \\ \varepsilon_{23} \\ \varepsilon_{24} \end{bmatrix}, \qquad \varepsilon_{ij} \sim i.i.d.\ N(0, \sigma^2)$$

Example 3

Parameter dimension $p = p_a + p_b$, no intercept, orthogonal subspaces.

Use vector-matrix notation,

$$Y = X\beta + \varepsilon, \qquad \varepsilon \sim N(0_n, \sigma^2 I_n)$$

The design matrix is

$$X = [X_A\ X_B]$$

where matrix X_A is $n \times p_a$, X_B is $n \times p_b$, and the two matrices are orthogonal to each other.

$$X_A^T X_B = 0_{p_a\, p_b}, \qquad X_B^T X_A = 0_{p_b\, p_a}$$

Then

$$\|Y\|^2 = \|P_A Y\|^2 + \|P_B Y\|^2 + \|(I - P_X)Y\|^2$$

which corresponds to the decomposition of sums of squares.

Example 4a — Orthogonal subspaces.

$$Y = X\beta + \varepsilon, \qquad \varepsilon \sim N(0_n, \sigma^2 I_n)$$

The design matrix is

$$X = [X_0\ X_A\ X_B\ X_C], \qquad C = A * B$$

where the matrices are pairwise orthogonal. Then there is a decomposition of sums of squares of subspaces.

$$\|Y\|^2 = \|P_0 Y\|^2 + \|P_A Y\|^2 + \|P_B Y\|^2 + \|P_C Y\|^2 + \|(I - P_X)Y\|^2$$

Example 4b — Orthogonal subspaces.

A concrete example: as in the previous example, the linear model is

$$Y = X\beta + \varepsilon, \qquad \varepsilon \sim N(0_n, \sigma^2 I_n), \qquad n = 12.$$

The design matrix is

$$X = [X_0\ X_A\ X_B\ X_C], \qquad C = A * B$$

where factor A has two levels, factor B has 3 levels, with 2 replicates for each combination.

With relabelling, the linear model can be written in the familiar 2-way ANOVA form:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \varepsilon_{ijk}, \quad \varepsilon_{ijk}\ i.i.d.\ \sim N(0, \sigma^2), \quad i = 1, 2;\ j = 1, 2, 3;\ k = 1, 2.$$

with parameter constraints

$$\sum_i \alpha_i = 0, \qquad \sum_j \beta_j = 0, \qquad \sum_i \gamma_{ij} = 0, \qquad \sum_j \gamma_{ij} = 0.$$

Now the model form

$$Y = X\beta + \varepsilon$$

can be reexpressed as

$$\begin{bmatrix} y_{111} \\ y_{112} \\ \hline y_{121} \\ y_{122} \\ \hline y_{131} \\ y_{132} \\ \hline y_{211} \\ y_{212} \\ \hline y_{221} \\ y_{222} \\ \hline y_{231} \\ y_{232} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ \hline 1 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 & 1 \\ \hline 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & -1 & -1 \\ \hline 1 & -1 & 1 & 0 & -1 & 0 \\ 1 & -1 & 1 & 0 & -1 & 0 \\ \hline 1 & -1 & 0 & 1 & 0 & -1 \\ 1 & -1 & 0 & 1 & 0 & -1 \\ \hline 1 & -1 & -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \mu \\ \alpha_1 \\ \beta_1 \\ \beta_2 \\ \gamma_{11} \\ \gamma_{12} \end{bmatrix} + \begin{bmatrix} \varepsilon_{111} \\ \varepsilon_{112} \\ \hline \varepsilon_{121} \\ \varepsilon_{122} \\ \hline \varepsilon_{131} \\ \varepsilon_{132} \\ \hline \varepsilon_{211} \\ \varepsilon_{212} \\ \hline \varepsilon_{221} \\ \varepsilon_{222} \\ \hline \varepsilon_{231} \\ \varepsilon_{232} \end{bmatrix}$$

Remarks

- This study is a balanced two-factor design.
- The design matrix contains four orthogonal subspaces with dimensions $(1, 1, 2, 2)$, respectively:
  - Subspace $C(X_0)$: spanned by column 1.
  - Subspace $C(X_A)$: spanned by column 2.
  - Subspace $C(X_B)$: spanned by columns 3 and 4.
  - Subspace $C(X_C)$: spanned by columns 5 and 6.

  Any two column vectors from two different subspaces are orthogonal to each other.

  In the above, $C(M)$, the span of matrix M, is also the column space of M consisting of all linear combinations of the column vectors of M.
- The pairwise orthogonality between subspaces implies that the ANOVA sums of squares

  $$\|Y\|^2 = \|P_0 Y\|^2 + \|P_A Y\|^2 + \|P_B Y\|^2 + \|P_C Y\|^2 + \|(I - P_X)Y\|^2$$

  are unaffected by the order of the factors A and B.
- If the design is unbalanced, say, by removing one row from the design matrix, then the pythagorean relationship in the SS decomposition no longer hold. Some adjustment is needed, such as the sequential ANONVA method.

(Related materials: Rice ch. 14; Casella and Berger ch. 11.)